Community Service Project

Report on

SOLID WASTE MANAGEMENT

Submitted to

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY KAKINADA

In partial fulfillment of the requirements

For the Award of the Degree of

Bachelor of Technology

in

Civil Engineering

By

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Under the Guidance of

Mrs.U.PALLAVI

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Department of Civil Engineering

SESHADRIRAO GUDLAVALLERU ENGINEERING COLLEGE

(An Autonomous Institute with Permanent Affiliation to JNTUK, Kakinada)

SESHADRIRAO KNOWLEDGE VILLAGE

GUDLAVALLERU - 521356

ANDHRA PRADESH

2026-2027

Program Book for
Community Service Report



Name of the college **:** Seshadri Rao Gudlavalleru Engineering College, Gudlavalleru, Krishna district.

Department **:** Department of Civil Engineering

Name of the Faculty Guide **:** Mrs.U.pallavi

DURATION OF THE CSP : FROM 27-06-2025 TO 13-08-2025.

NAME OF THE STUDENTS : V. UMA SRI(24481A0156)

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PROGRAM OF STUDY : Solid waste management

YEAR OF STUDY : III B.Tech, I -Sem.

DATE OF SUBMSSION : 13-11-2025.

Student's Declaration

We are a students of community service program of the Department of Civil Engineering, Seshadri Rao Gudlavalleru Engineering College do hereby individually declare that we have completed the mandatory community service project from **27-06-2025** to 13-08-2025 in Kaikaluru village of Eluru District under the Faculty Guidance of Mrs.U.pallavi, Department of Civil Engineering in Seshadri Rao Gudlavalleru Engineering College, Gudlavalleru.

(Student signature)

Endorsements:

Faculty Guide :

Head of the Department :

Certificate from Official of the Community

This is to certify that the following students of the Department of Civil Engineering, Seshadri Rao Gudlavalleru Engineering College underwent community service project in Kaikaluru, Eluru District from 27-06-2025 to 13-08-2025.V. UMA SRI (24481A0156), G. CHAKRI SAI (24481A0115), K. KRISHNA KOWSHIK (24481A0122)

The overall performance of the Community Service Volunteers during theircommunity service is found to be satisfactory.

Authorized Signature with Date and Seal

ACKNOWLEDGEMENT

We would like to express our gratitude and sincere thanks to **Mrs.U.pallavi,** Assistant Professor, Department of Civil Engineering, for her constant guidance, supervision and motivation in completing the Community Service Project (CSP).

We are very much indebted to Dr. A.H.L. SWAROOP, Head of the Department of Civil Engineering, for his valuable suggestions and encouragement throughout our project.

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We wish to thank all the community people who helped in to do project in successfulway.

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We also wish to express my indebtedness to my parents as well as my family memberswhose blessings and support always helped me to face the challenges ahead.

Project Associates

V. UMA SRI (24481A0156)
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ABSTRACT

Solid waste management is an essential environmental practice that helps maintain cleanliness, protect public health, and promote sustainable living. In rural and semi-urban areas, solid waste is generated from households, shops, public places, and other daily activities. Improper collection, mixing, and disposal of waste can result in pollution, blocked drainage systems, unpleasant surroundings, and loss of recyclable materials. Effective management of solid waste requires proper segregation, collection, processing, recycling, and safe disposal.

This Community Service Project was conducted in and around Kaikaluru village to study the existing conditions of solid waste generation, collection, segregation, and disposal. The project involved field observations and interactions with local residents and shop owners to understand the common types of waste generated and the existing waste-handling practices. Particular attention was given to the separation of biodegradable and non-biodegradable waste and the possible recovery of recyclable materials.

The study focuses on identifying simple and practical methods for improving solid waste management at the community level. Suitable measures such as source segregation, separate collection, composting of biodegradable waste, reuse and recycling of recyclable materials, and proper disposal of residual waste were considered. These practices can help reduce the amount of waste sent for final disposal and improve the overall cleanliness of the village.

The project highlights the importance of creating awareness about proper source segregation management approach in Kaikaluru village. Proper handling of waste at its source, efficient collection, and better utilization of recyclable and biodegradable materials can contribute to a cleaner, healthier, and more sustainable community.

CHAPTER 1
EXECUTIVE SUMMARY

This Community Service Project focuses on studying the existing solid waste management practices in and around Kaikaluru village. Solid waste generated from households, shops, and public places includes biodegradable waste, plastic, paper, glass, metal, and other discarded materials. Improper handling and disposal of solid waste can cause environmental pollution, unpleasant surroundings, blocked drainage systems, and other community-level problems. Proper management of solid waste is essential for maintaining a clean, healthy, and sustainable environment.

The project involved direct observation, field visits, collection of information, and interaction with local residents and shop owners to understand the existing practices related to waste generation, segregation, collection, and disposal. The study focused on identifying common types of waste generated in the selected area and understanding the problems associated with improper waste handling. Suitable and practical approaches for improving solid waste management at the community level were considered based on the observations.

Learning Objectives:

- To understand the existing solid waste management practices in Kaikauru village.
- To identify and classify the major types of solid waste generated in the selected area.
- To study the existing methods of waste collection, segregation, and disposal.
- To understand the environmental and public health problems associated with improper waste management.
- To create awareness about proper segregation of wet and dry waste at the community level.
- To enhance communication, field observation, data collection, and teamwork skills.

Learning Outcomes:

- I have improved my communication skills through interaction with local residents and shop owners.
- I can identify and classify different types of solid waste generated in the community.
- I can recognize solid waste management problems as important environmental and public health challenges.
- I have gained practical knowledge of waste segregation, collection, and disposal practices

CHAPTER 2
SOLID WASTE MANAGEMENT

- Solid waste management is an important environmental and public health concern in both urban and rural areas. Solid waste is generated from households, shops, institutions, agricultural activities, and other daily activities. It includes materials such as food waste, paper, plastic, glass, metal, cloth, and other discarded materials. When solid waste is not properly managed, it can accumulate in open areas and create unpleasant surroundings, foul odour, and conditions that may affect public health and the environment.

- The quantity of solid waste generated in communities has increased due to population growth, changing lifestyles, increased consumption, and the widespread use of packaged products. Rural areas also face difficulties in managing waste because of limited collection facilities, inadequate disposal systems, and improper segregation practices. Therefore, proper solid waste management is necessary to maintain clean surroundings and reduce environmental pollution.

- Solid waste management involves a series of activities including waste generation, segregation, storage, collection, transportation, recycling, treatment, and final disposal. These activities should be carried out systematically so that waste can be managed safely and efficiently. Among these activities, segregation at the source is one of the most important steps because it helps to separate different types of waste before collection and disposal.

2.1 TYPES OF SOLID WASTE

- Solid waste can be classified according to its nature and the way it can be managed. The major types commonly found in communities are biodegradable, recyclable, non-recyclable, and special waste.

TYPE OF SOLID WASTE	EXAMPLES	MANAGEMENT METHOD
Biodegradable	Food waste, vegetable waste, leaves	Composting or biological treatment
Recyclable	Paper, cardboard, glass, metals, recyclable plastics	Recycling
Non-recyclable	Mixed packaging and contaminated materials	Appropriate disposal
Special waste	Batteries, chemical containers, e-waste	Separate and safe handling

- Biodegradable waste consists of materials that can naturally decompose through biological processes. Food scraps, vegetable and fruit waste, leaves, and garden waste are common

examples. Such waste can be treated through composting or other suitable biological methods.

- Recyclable waste includes materials that can be collected and processed to produce useful products again. Paper, cardboard, glass, metals, and certain plastics are common examples. Proper segregation helps in recovering these materials and reduces the quantity of waste sent for final disposal.

- Non-recyclable waste consists of materials that cannot be easily recycled using commonly available facilities. Certain contaminated packaging materials and mixed wastes may fall under this category. Such waste should be collected and disposed of through appropriate waste management systems.

- Special waste includes materials that require separate handling because of their nature. Batteries, chemical containers, sanitary waste, and electronic waste should not be mixed with ordinary household waste and should be handled through suitable disposal systems.

• 2.2 WASTE SEGREGATION

- Waste segregation means separating waste into different categories at the place where it is generated. It is an essential step in an effective solid waste management system. Segregation at source reduces the mixing of biodegradable, recyclable, and other wastes and makes further processing easier.

- A simple segregation system can separate waste into biodegradable and non-biodegradable categories. Biodegradable waste can be treated through composting or other suitable methods, while recyclable materials can be collected separately for recycling. Waste that cannot be recovered or recycled can be sent for appropriate final disposal.

- Proper segregation reduces the amount of waste reaching dumping areas. It improves the quality of recyclable materials and reduces the effort required for sorting after collection. Therefore, source-level segregation is an important practice for maintaining a clean and sustainable community.

• 2.3 COLLECTION AND STORAGE OF SOLID WASTE

- After segregation, waste should be stored safely until it is collected. Waste storage containers should be suitable for the type and quantity of waste generated. Containers should be kept in accessible and hygienic locations to prevent littering and scattering of waste.

- Regular collection is necessary to prevent the accumulation of waste in residential areas, streets, and public places. Collection systems should be planned according to the quantity of

waste generated and the requirements of the community. Proper collection and transportation reduce the possibility of waste being dumped in open areas or disposed of into drains and water bodies.

- In rural communities, local participation is particularly important for maintaining regular waste collection and preventing improper disposal. A simple and organized collection system can improve cleanliness and make waste management more effective.

- **2.4 RECYCLING AND REUSE**
- Recycling is the process of recovering useful materials from waste and converting them into new products. Materials such as paper, cardboard, glass, metals, and recyclable plastics can be processed and reused. Recycling reduces the demand for fresh raw materials and helps conserve natural resources.

- Reuse is another important method of reducing waste. Materials that are still useful can be used again instead of being immediately discarded. Reusing suitable containers, bags, and other materials can reduce the amount of waste generated.

- Recycling and reuse also help reduce the volume of waste requiring final disposal. When recyclable materials are properly separated at the source, they can be collected more efficiently and transferred to appropriate recycling facilities.

- **2.5 ENVIRONMENTAL AND HEALTH IMPACTS**
- Improper disposal of solid waste can create several environmental problems. Open dumping of waste can contaminate soil and water and may produce unpleasant odours. Waste thrown into drains can obstruct the flow of water and contribute to waterlogging during rainfall.

- Accumulated waste can attract insects, rodents, and other organisms that may contribute to the spread of diseases. Burning waste in open areas can release smoke and harmful pollutants into the atmosphere. Plastic and other non-biodegradable materials can remain in the environment for long periods and cause further pollution.

- Improper waste management also affects the cleanliness and appearance of villages and public spaces. Therefore, systematic collection, segregation, transportation, treatment, and disposal are necessary to reduce environmental and health-related problems.

2.6 CHALLENGES IN SOLID WASTE MANAGEMEN

- Several challenges affect effective solid waste management in rural communities. One major challenge is the mixing of different types of waste at the source. When biodegradable and

recyclable materials are mixed with other waste, recovery and recycling become more difficult.

- Other challenges include inadequate collection facilities, irregular transportation, limited disposal infrastructure, lack of suitable storage containers, and insufficient community participation. Limited resources and improper disposal practices can further increase the problem.

- Addressing these challenges requires cooperation between local authorities and community members. Proper planning, regular collection, source segregation, suitable storage facilities, and responsible disposal practices can improve the overall waste management system**.**

• 2.7 SUSTAINABLE SOLID WASTE MANAGEMENT

- Sustainable solid waste management focuses on reducing waste generation and managing the remaining waste in an environmentally responsible manner. The basic approach follows the principles of Reduce, Reuse, and Recycle.

- Reducing waste means avoiding unnecessary consumption and minimizing the use of materials that quickly become waste. Reusing means using suitable materials more than once instead of discarding them after a single use. Recycling involves recovering useful materials and processing them into new products.

- Community participation is essential for achieving sustainable waste management. Households, local authorities, educational institutions, shops, and other community groups can contribute by following proper segregation, storage, collection, and disposal practices.

• 2.8 ROLE OF COMMUNITY PARTICIPATION

- Effective solid waste management cannot be achieved only through collection and disposal facilities. The participation of local residents is equally important. Every household and community member has a role in reducing waste, segregating it properly, avoiding littering, and following appropriate disposal practices.

- Educational institutions can also contribute by encouraging students to follow responsible waste management practices and participate in community-based environmental activities. Cooperation between residents and local authorities can improve the cleanliness of public spaces and support the long-term sustainability of waste management practices.

CHAPTER 3

SOLID WASTE MANAGEMENT RULES AND FIELD ANALYSIS

- The management of solid waste in India is guided by regulations that aim to ensure proper segregation, collection, transportation, processing, and disposal of waste. The latest framework is the Solid Waste Management Rules, 2026, notified by the Ministry of Environment, Forest and Climate Change. These rules came into force on 1 April 2026 and apply to both urban and rural local bodies. They provide a systematic approach for managing solid waste and reducing its impact on the environment and public health.

- The rules cover domestic, institutional, commercial, and other non-residential solid waste generators. They emphasize responsible waste management by waste generators and local authorities. The framework aims to ensure that waste is properly managed from the point of generation until its final treatment or disposal. Certain wastes such as biomedical waste, e-waste, battery waste, hazardous waste, and radioactive waste are covered under separate regulations.

3.1 SOLID WASTE MANAGEMENT RULES

- One of the important principles of solid waste management is segregation at source. Waste should be separated according to its nature before it is collected. Segregation helps prevent different types of waste from getting mixed and makes recycling, composting, treatment, and disposal easier.

- The rules also emphasize the proper storage and collection of waste. Waste generators are expected to follow the prescribed methods for handling waste, while local bodies are responsible for organizing suitable systems for collection, transportation, processing, and disposal. Proper management at each stage helps reduce open dumping and environmental pollution.

- Biodegradable waste can be processed through suitable biological methods such as composting. Recyclable materials can be recovered and sent for recycling, while other waste should be treated or disposed of through appropriate facilities. This approach helps reduce the quantity of waste reaching disposal sites and supports the recovery of useful materials.

- The rules also encourage responsible participation by households, institutions, commercial establishments, and local communities. Proper waste management cannot be achieved only through government facilities. The cooperation of waste generators is necessary for maintaining cleanliness and ensuring that waste is handled correctly.

- The implementation of solid waste management rules at the village level is particularly important

because rural areas may have limited collection, transportation, and processing facilities. A simple and organized system involving source segregation, proper storage, regular collection, and suitable disposal can improve environmental conditions and reduce waste-related problems.

3.2 CHALLENGES IN SOLID WASTE MANAGEMENT IN KAIKALURU VILLAGE

During the community service project activities in Kaikaluru village, the importance of proper solid waste management was considered with respect to the local surroundings and common waste-handling practices. Several general challenges can affect effective solid waste management in rural communities.

1. Improper Segregation:
One of the major challenges is the mixing of biodegradable and non-biodegradable waste. When different types of waste are mixed together, recyclable materials become difficult to recover and organic waste becomes unsuitable for easy biological treatment.

2.Inadequate Waste Storage:
The absence of suitable and clearly separated waste storage facilities can result in waste being placed in open areas. Proper storage at the source is necessary to prevent littering and to make collection easier.

3.Irregular Collection:
Regular collection is important for preventing the accumulation of waste. Where collection systems are insufficient or irregular, residents may dispose of waste in open spaces, roadside areas, or other unsuitable locations.

4.Lack of Proper Disposal Practices:
Improper disposal can cause environmental pollution and affect the cleanliness of the village. Waste dumped near roads, drains, or water bodies may create unpleasant surroundings and can contribute to blockage and water pollution.

5.Limited Community Participation:
The success of waste management depends on the participation of residents. If households do not separate waste or follow proper disposal practices, the collection and processing system becomes less effective.

6.Limited Waste Management Facilities:
Rural areas may have limited access to recycling, composting, and other waste-processing facilities. This can make it difficult to manage different types of waste after collection.

7.Lack of Continuous Monitoring:

Waste management requires regular observation and monitoring. Without continuous checking, improper disposal practices may continue and waste can again accumulate in public areas.

3.3 ANALYSIS OF SOLID WASTE MANAGEMENT PRACTICES

- The analysis of solid waste management focuses on understanding how waste is generated, stored, segregated, collected, and disposed of within the community. The major purpose of the analysis is to identify common problems and understand the need for an organized waste management system.

- The first important observation is that source segregation plays a major role in improving waste management. When biodegradable and non-biodegradable waste are kept separately, the further handling of waste becomes easier. Organic waste can be considered for composting, while recyclable materials can be recovered separately.

- Another important aspect is the availability of suitable storage facilities. Waste should be stored in a manner that prevents scattering, littering, and mixing of different waste categories. Proper storage also helps collection workers handle waste more safely and efficiently.

- Collection and transportation are also important parts of the waste management process. Even when waste is segregated properly at the source, irregular collection can result in accumulation. Therefore, a suitable collection system is necessary to maintain cleanliness in residential and public areas.

- The analysis also shows the importance of reducing waste generation. Households can reduce waste by avoiding unnecessary disposable materials, reusing suitable items, and separating recyclable materials. These practices can reduce the quantity of waste requiring final disposal.

3.3.1 KEY OBSERVATIONS

- The following key observations are relevant to solid waste management at the community level:

- - Different types of solid waste are generated from households and other daily activities.
- - Mixing biodegradable and non-biodegradable waste makes further waste processing difficult.
- - Proper segregation at the source can improve the handling of waste.
- - Suitable storage facilities are necessary to prevent littering and scattering.
- - Regular collection and transportation are important for maintaining clean surroundings.
- - Recyclable materials should be separated so that they can be recovered and reused.
- - Organic waste can be considered for suitable biological treatment such as composting.
- - Community participation is necessary for maintaining proper waste management practices.
- - Continuous monitoring can help identify and reduce improper disposal practices.

- - Cooperation between residents and local authorities is important for sustainable waste management.

3.3.2 OVERALL ANALYSIS

- The overall analysis indicates that effective solid waste management requires a systematic approach from the point of waste generation to final disposal. Source segregation is a basic and important step because it prevents different types of waste from being mixed and improves the possibility of recycling and treatment.

- Proper storage, regular collection, suitable transportation, recycling, biological treatment, and safe final disposal should work together as parts of one system. Improving only one stage may not be sufficient if the remaining stages are not properly managed.

- Community participation is equally important. Households and other waste generators can contribute by reducing waste generation, separating waste at the source, avoiding littering, and following proper disposal practices. Local authorities can support these efforts by providing suitable collection and waste-processing facilities.

- Therefore, a simple, organized, and continuous solid waste management system can help improve cleanliness, reduce environmental pollution, conserve useful resources, and create healthier surroundings in Kaikaluru village. The principles of segregation, reduction, reuse, recycling, proper collection, and safe disposal provide a practical foundation for sustainable community-level waste management.

CHAPTER 4

METHODS FOR SOLID WASTE COLLECTION AND MANAGEMENT

Solid waste management requires proper handling of waste from the point of generation to its final disposal. In Kaikaluru village, proper waste management can be achieved through simple and systematic practices such as segregation, collection, transportation, recycling, treatment, and safe disposal. These methods help to maintain cleanliness and reduce environmental pollution.

4.1 METHODS FOR SOLID WASTE COLLECTION

1. Segregation at Source

Waste should be separated at the place where it is generated. Biodegradable waste such as food waste, vegetable waste, and leaves should be kept separately from non-biodegradable waste such as plastic, paper, glass, and metals. Source segregation makes further processing and recycling easier.

2. Separate Storage

The segregated waste should be stored in suitable containers until collection. Proper storage prevents waste from being scattered and reduces littering in residential and public areas. The storage containers should be placed at convenient and accessible locations.

3. Collection of Waste

Waste should be collected regularly from houses and designated collection points. Regular collection prevents the accumulation of waste and helps maintain clean surroundings. During collection, different types of waste should be kept separate as far as possible.

4. Transportation

After collection, the waste should be transported safely to the appropriate processing or disposal location. Proper transportation prevents waste from falling or scattering on roads and other public areas.

5. Recycling and Reuse

Materials such as paper, cardboard, glass, metals, and suitable plastics can be separated and sent

for recycling. Items that can still be used should be reused instead of being discarded. Recycling and reuse reduce the quantity of waste requiring final disposal.

6. Treatment of Biodegradable Waste

Biodegradable waste such as food waste, vegetable waste, and leaves can be treated by methods such as composting. Composting converts organic waste into useful compost and reduces the amount of waste sent for disposal.

7. Safe Disposal

Waste that cannot be reused, recycled, or treated should be disposed of through suitable authorized waste management systems. Open dumping and open burning should be avoided because they can cause environmental pollution.

4.2 SOLID WASTE MANAGEMENT TECHNIQUES

- Effective solid waste management includes reducing waste generation, segregation, collection, transportation, recycling, treatment, and final disposal.

- Waste Reduction: Unnecessary use of disposable materials should be avoided, and reusable products should be preferred wherever possible.

- Waste Segregation: Different types of waste should be separated at the source. This makes recycling and treatment easier and reduces the problems caused by mixed waste.

- Waste Collection and Transportation: Waste should be collected regularly and transported safely to appropriate processing or disposal locations.

- Recycling and Reuse: Recyclable materials should be recovered and sent for recycling, while reusable materials should be used again whenever possible.

- Composting: Biodegradable waste can be converted into compost through suitable composting methods. This reduces the quantity of organic waste requiring disposal.

- Final Disposal: Waste that cannot be recovered or treated should be disposed of safely through appropriate facilities. Open dumping and burning should be avoided.

4.3 IMPORTANCE OF PROPER WASTE MANAGEMENT

- Proper solid waste management helps maintain clean and healthy surroundings in Kaikaluru village. Regular collection prevents waste accumulation in open areas, roads, and drains. Segregation at source improves recycling and treatment efficiency.

- Proper waste management also reduces soil, water, and air pollution and helps prevent the spread of waste-related health problems. Community members, waste collection workers, and local authorities should work together to maintain an effective waste management system.

- Community participation is important for the success of the waste management process. Residents can contribute by reducing waste generation, segregating waste at source, avoiding littering, reusing useful materials, and following proper disposal practices. Through these simple practices, Kaikaluru village can move towards a cleaner and more sustainable environment.

CHAPTER 5
COMMUNITY SERVICE PART

The Community Service Project focused on studying solid waste management practices in Kaikaluru village and identifying the common problems related to the collection, segregation, storage, and disposal of solid waste. The main objective of the project was to understand the existing solid waste management practices and create awareness about proper waste segregation.

The project activities were carried out through field visits and observations in different areas of Kaikaluru village. The types of solid waste generated and the existing methods of collection and disposal were observed. Particular attention was given to the mixing of biodegradable and non-biodegradable waste.

The field activities helped in identifying the need for proper segregation of solid waste at the source. Based on observations, the existing two-bin system was considered practical for separating biodegradable and non-biodegradable waste.

Field Work and Activities:

Throughout the 8-week project, several field activities were carried out:

- Week 1–2: Field visits to observe solid waste collection and disposal practices in Kaikaluru village.

- Week 3–4: Identification and observation of different types of solid waste and segregation problems.

- Week 5~~6~~: Awareness about proper use of the existing two-bin system for separating biodegradable and non-biodegradable waste.

- Week 7: Observation of the existing two-bin system and its practical use.

- Week 8: Review of the project activities and documentation of the results.

CONCLUSION:

The Community Service Project provided practical experience in understanding solid waste management practices in Kaikaluru village. The field activities helped in identifying the problems related to improper segregation and disposal of solid waste.

The project focused on awareness about proper use of the existing system for biodegradable and non-biodegradable waste. The project helped demonstrate the importance of source segregation and provided practical experience in applying a simple solid waste management method in the community.

ACTIVITY LOG FOR THE FIRST WEEK (FROM 27-06-2025 TO 03-07-2025_)

DAY & DATE	BRIEF DESCRIPTION OF THE DAILYACTIVITY	LEARNING OUTCOME	Person In-charge Signature
Day –1 & 27-06-2025	Today we received an introduction and detailed briefing about our CSP project on Solid Waste Management. Our guide explained the project objectives and the activities to be carried out during the field work.	I understood the overall project objectives and got a clear idea about the Week 1–2 plan.	
Day –2 & 28-06-2025	We visited nearby residential areas in Kaikaluru village and observed how households store and dispose of their solid waste. We also observed the common types of waste generated in the area.	I learned how solid waste is generated and handled in residential areas.	
Day –3 & 30-06-2025	We observed the existing solid waste collection practices in the selected areas and noted how waste was collected and stored before disposal.	I gained knowledge about the existing solid waste collection practices.	
Day –4 & 01-07-2025	We visited common waste disposal areas and observed how solid waste was disposed of. We also noted the problems caused by mixing different types of waste.	I understood the existing disposal practices and problems related to mixed waste.	
Day –5 & 02-07-2025	We collected photographs, notes, and observations from the visited locations and recorded the information for further project work.	I learned how to document and record field observations properly.	
Day –6 & 03-07-2025	We reviewed all the collected observations as a team and discussed the common solid waste management problems found during the field visits.	I improved my understanding of the identified solid waste management problems.	

WEEKLY REPORT
WEEK – 1 (FROM 27-06-2025 TO 03-07-2025_)

Objective of the Activity Done:

- To understand the existing solid waste collection and disposal practices in Kaikaluru village and observe the common waste management problems.

Detailed Report:

- During Week 1, we mainly focused on understanding the project work and observing the actual solid waste management conditions in the community. We visited residential areas and other selected locations in Kaikaluru village.

- We observed how solid waste was collected, stored, and disposed of in different areas. We also identified different types of waste and observed the problems caused by improper segregation and disposal.

- We collected photographs and notes from the visited locations, which helped us understand the existing solid waste management practices clearly.

- This week helped us gain a clear understanding of the field conditions and prepare for the upcoming activities of the project.

ACTIVITY LOG FOR THE SECOND WEEK(FROM 09-07-2025 TO 04-07-2025)

DAY & DATE	BRIEF DESCRIPTION OF THE DAILY ACTIVITY	LEARNING OUTCOME	Person In-charge Signature
Day –7 & 04-07-2025	We visited residential areas in Kaikaluru village and observed how people collect and store solid waste in their homes.	I learned about household solid waste handling practices.	
Day –8 & 05-07-2025	We interacted with local people and observed how biodegradable and non-biodegradable waste is disposed of.	I understood the common waste disposal practices.	
Day –9 & 06-07-2025	We visited different areas and observed the places where solid waste is commonly collected and disposed of.	I learned about common waste collection and disposal locations.	
Day –10 & 07-07-2025	We observed the problems caused by mixing different types of solid waste during collection and disposal.	I understood the difficulties caused by mixed waste.	
Day –11 & (08-07-2025	We identified and categorized the commonly generated waste into biodegradable and non-biodegradable types.	I gained practical knowledge of waste categorization.	
Day –12 & 09-07-2025	We categorized the collected waste into recyclable and non-recyclable types and recorded the quantities.	I developed data analysis and report-writing skills.	

WEEKLY REPORT
WEEK – 2(FROM 09-07-2025 TO 04-07-2025)

Objective of the Activity Done:

To study solid waste collection, storage, segregation, and disposal practices in residential and other selected areas of Kaikaluru village.

Detailed Report:

In Week 2, we continued our field work in Kaikaluru village and observed solid waste management practices in different areas. We visited residential areas and interacted with local people to understand how solid waste is collected, stored, and disposed of.

We observed the different types of solid waste generated in the selected areas and identified biodegradable and non-biodegradable waste. We also observed whether the waste was being separated or mixed during collection and disposal.

We visited common waste collection and disposal areas and noted the problems caused by mixing different types of waste. Photographs and field observations were collected to understand the existing practices clearly.

The collected information helped us identify the major solid waste management problems in the village. This week gave us a better understanding of waste segregation and helped us prepare for the next stage of the project.

ACTIVITY LOG FOR THE THIRD WEEK(FROM 11-07-2025 TO 16-07-2025)

DAY & DATE	BRIE DESCRIPTION OF THE DAILY ACTIVITY	LEARNING OUTCOME	Person In-charge Signature
Day –13 & 11-07-2025	We visited selected areas in Kaikaluru village and observed the different types of solid waste generated in the surroundings.	I learned to identify different types of solid waste.	
Day –14 & 12-07-2025	We observed biodegradable and non-biodegradable waste and noted how they were stored and handled in different areas.	I understood the difference between biodegradable and non-biodegradable waste.	
Day –15 & 13-07-2025	We observed places where different types of waste were mixed together and recorded the segregation problems.	I learned about the problems caused by mixed waste	
Day –16 & 14-07-2025	We visited residential and other selected areas to observe the existing waste segregation practices.	I understood the existing waste segregation practices.	
Day –17 & 15-07-2025	We prepared notes on the major waste segregation problems based on our field observations.	I improved my problem identification and observation skills.	
Day –18 & 16-07-2025	We reviewed the collected information and photographs and prepared the observations for the Week 3 report.	I improved my documentation and report-writing skills.	

WEEKLY REPORT
WEEK -3 (FROM 11-07-2025 TO 16-07-2025)

Objective of the Activity Done:

To identify different types of solid waste and observe the existing waste segregation problems in selected areas of Kaikaluru village.

Detailed Report:

During Week 3, we focused on identifying different types of solid waste and observing the existing segregation practices in Kaikaluru village. We visited selected residential and other areas to understand the types of waste commonly generated in the community.

We observed biodegradable and non-biodegradable waste in different locations and noted how the waste was stored and handled. We also observed places where different types of waste were mixed together.

The field observations helped us identify the common problems related to improper segregation. Mixed waste makes the further collection, recycling, treatment, and disposal of waste more difficult.

Overall, Week 3 helped us gain practical knowledge about different types of solid waste and the problems associated with improper segregation. The observations made during this week were useful for planning the next activities of the project.

ACTIVITY LOG FOR THE FOURTH WEEK(FROM 18-07-2025 TO 23-07-2025)

DAY & DATE	BRIEF DESCRIPTION OF THE DAILYACTIVITY	LEARNING OUTCOME	Person In-charge Signature
Day –19 & 18-07-2025	We visited selected areas in Kaikaluru village and observed the places where solid waste was commonly mixed and stored.	I understood the common locations where waste segregation problems occur.	
Day –20 & 19-07-2025	We observed biodegradable and non-biodegradable waste in different areas and recorded the common waste materials found there.	I learned to identify and categorize different types of waste.	
Day –21 & 20-07-2025	We observed the problems caused by mixing wet and dry waste during collection and storage.	I understood how mixed waste creates difficulties in proper waste management.	
Day –22 & 21-07-2025	We visited residential areas and observed the existing methods used for storing and separating solid waste.	I gained practical knowledge of household waste segregation practices.	
Day –23 & 22-07-2025	We reviewed the field observations and identified the main problems related to improper waste segregation.	I improved my problem-identification and observation skills.	
Day –24 & 23-07-2025	We organized the collected photographs and notes and prepared the information for the Week 4 report.	I improved my documentation and report-writing skills	

WEEKLY REPORT
WEEK – 4 (FROM 18-07-2025 TO 23-07-2025)

Objective of the Activity Done:

To observe the existing waste segregation practices and identify the common problems caused by mixing different types of solid waste in Kaikaluru village.

Detailed Report:

During Week 4, we continued our field work in selected areas of Kaikaluru village and focused on observing the existing waste segregation and storage practices. We visited residential and other selected locations to understand how different types of solid waste were handled.

We observed biodegradable and non-biodegradable waste in different areas and noted the common materials found in the waste. We also observed places where different types of waste were mixed together during storage and collection.

The field observations helped us understand the problems caused by improper segregation. Mixing wet and dry waste makes proper collection, recycling, treatment, and disposal more difficult.

Overall, Week 4 helped us gain practical knowledge about waste segregation practices and the problems associated with mixed waste. The information collected during Weeks 3 and 4 helped us plan the next stage of the Community Service Project.

ACTIVITY LOG FOR THE FIFTH WEEK(FROM 25-07-2025 TO 30-07-2025)

DAY & DATE	BRIEF DESCRIPTION OF THE DAILYACTIVITY	LEARNING OUTCOME	Person In-charge Signature
Day –25 & 25-07-2025	We reviewed the segregation problems identified during the previous weeks and discussed a suitable method for improving waste segregation.	I understood the need for a simple waste segregation method.	
Day –26 & 26-07-2025	We discussed the use of separate bins for collecting biodegradable and non-biodegradable waste.	I learned how separate bins can help in source segregation.	
Day –27 & 27-07-2025	We discussed proper use of the exist two-bin system for different types of solid waste.	I understood the basic planning required for source segregation.	
Day –28 & 28-07-2025	We identified awareness needs and proper use of the existing system.	We identified suitable locations and requirements for placing the two-bin system.	
Day –29 & 29-07-2025	We discussed the existing bins and identified the type of waste to be placed in each bin.	I gained practical knowledge about promoting waste segregation system.	
Day –30 & 30-07-2025	We reviewed the existing system and discussed the activities to be carried out during the awareness stage.	I understood the next steps for promoting the two-bin system.	

WEEKLY REPORT
WEEK – 5 (FROM 18-07-2025 TO 23-07-2025)

Objective of the Activity Done:

To create awareness about proper use of the existing system for separating wet and dry waste based on problems identified during previous weeks.

Detailed Report:

- During Week 5, we continued the project work based on the waste segregation problems identified during the previous weeks. We discussed the need for a simple and practical method to separate different types of solid waste at the source.

•Based on observations, we focused on awareness about separating wet and dry waste. We discussed the purpose of each bin and the types of waste. That waste would be placed in them.

•We discussed proper use of the existing two-bin system. The system was already available and can be easily understood and used for waste segregation.

•Overall, Week 5 helped us move from identifying segregation problems to planning awareness activities. The work carried out during this week provided the basis for the awareness activities in Week 6.

ACTIVITY LOG FOR THE SIXTH WEEK(FROM 01-08-2025 TO 06-08-2025)

DAY & DATE	BRIEF DESCRIPTION OF THE DAILYACTIVITY	LEARNING OUTCOME	Person In-charge Signature
Day –31 & 01-08-2025	We reviewed the existing two-bin system and discussed the proper use of the awareness activities.	I understood the steps required for proper use of the system.	
Day –32 & 02-08-2025	We observed the existing bins at the selected location and collecting different types of solid waste.	I learned proper source-segregation system practically.	
Day –33 & 03-08-2025	We identified the first bin for biodegradable waste and the second bin for non-biodegradable waste.	I understood the proper classification of waste for each bin.	
Day –34 & 04-08-2025	We demonstrated the separation of biodegradable and non-biodegradable waste using the two-bin system.	I gained practical knowledge of source segregation	
Day –35 & 05-08-2025	We observed the use and arrangement of the two-bin system and made necessary observations.	I learned how to observe the practical functioning of the system	
Day –36 & 06-08-2025	We reviewed the awareness work and recorded the observations for the project report.	I improved my documentation and evaluation skills.	

WEEKLY REPORT
WEEK – 6 (FROM 01-08-2025 TO 06-08-2025)

Objective of the Activity Done:

To create awareness about proper use of the existing system for separating solid waste at the source.

Detailed Report:

- During Week 6, we continued the project work based on observations and focused on awareness about proper use of the existing system as a practical method for separating different types of solid waste.

- The existing bins were observed at the selected location. One bin was designated for biodegradable waste, while the other was designated for non-biodegradable waste. The arrangement was made simple so that the different types of waste could be separated easily.

- We carried out awareness activities and observed the separation of biodegradable and non-biodegradable waste. This helped us understand how source segregation can be carried out using a simple method.

- Overall, Week 6 focused on awareness and proper use of the existing system. The work carried out during this week provided a basis for observing and reviewing the practical use of the system in Week 7.

ACTIVITY LOG FOR THE SEVENTH WEEK (FROM 08-08-2025TO13-08-2025)

DAY & DATE	BREIF DESCRIPTION OF THE DAILYACTIVITY	LEARNING OUTCOME	Person In-charge Signature
Day –37 & 08-08-2025	We visited the existing two-bin system and observed its condition and use after observation.	I learned how to monitor an proper waste segregation system.	
Day –38 & 09-08-2025	We observed the waste placed in the two bins and checked whether biodegradable and non-biodegradable waste were kept separately.	I understood the practical use of source segregation.	
Day –39 & 10-08-2025	We reviewed the handling and storage of waste in the two-bin system and recorded the observations.	I learned the importance of maintaining proper waste separation.	
Day –40 & 11-08-2025	We observed the practical working of the two-bin system and identified any difficulties in its use.	I gained practical knowledge about the working of the system.	
Day –41 & 12-08-2025	We discussed the observations collected during awareness activities and reviewed the usefulness of the system.	I learned how to review the results of a practical activity.	
Day –42 & 13-08-2025	We documented the observations and photographs from the existing two-bin system for the project report.	I improved my documentation and reporting skills.	

WEEKLY REPORT
WEEK – 7 (FROM 08-08-2025TO13-08-2025)

Objective of the Activity Done:

To observe the existing two-bin system and review proper segregation biodegradable and non-biodegradable waste.

Detailed Report:

•During Week 7, we continued the project work by observing the existing system in Week 6. The main focus of this week was to observe the practical use and condition of the existing system.

- We visited the selected location and observed the two-bin arrangement. The waste placed in the bins was checked to understand whether biodegradable and non-biodegradable waste were being kept separately.

- We also observed the handling and storage of waste in the two-bin system and recorded the important observations. This helped us understand the practical functioning of source segregation.

•Overall, Week 7 helped us observe the existing system and review segregation practices. The observations made during this week provided the basis for the final review and documentation activities planned for Week 8.

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ACTIVITY LOG FOR THE EIGHTH WEEK(FROM 15-08-2025 TO 20-08-2025)

DAY & DATE	BRIEF DESCRIPTION OF THE DAILYACTIVITY	LEARNING OUTCOME	Person In-charge Signature
Day –43 & 15-08-2025	We reviewed the observations collected during the previous week regarding the use of the two-bin system.	I understood the overall observations from the existing system.	
Day –44 & 16-08-2025	We organized the photographs, field notes, and other information collected throughout the project.	I learned how to organize project records systematicall	
Day –45 & 17-08-2025	We reviewed the activities carried out from the beginning of the project and discussed the progress achieved.	I understood the complete process followed during the project.	
Day –46 & 18-08-2025	We reviewed the practical outcome of awareness segregation system and discussed its usefulness.	I learned how to evaluate the outcome of a practical activity.	
Day –47 & 19-08-2025	We prepared the final observations and organized the information required for the project report.	I improved my report preparation and documentation skills	
Day –48 & 20-08-2025	We completed the final review of the Community Service Project and discussed the overall learning from the field work.	I gained overall practical knowledge of solid waste management	

WEEKLY REPORT
WEEK – 8 (FROM 15-08-2025 TO 20-08-2025.)

Objective of the Activity Done:

To review the completed project activities, document the observations, and summarize the practical outcome of waste segregation awareness.

Detailed Report:

During Week 8, we continued the project work based on the observations made during Week 7. The main focus of this week was to review the complete project activities and organize the information collected during the field work.

We reviewed the observations related to the two-bin system and discussed its practical use for separating biodegradable and non-biodegradable waste. The photographs, notes, and other field information collected throughout the project were organized for documentation.

We also reviewed the activities carried out from the beginning of the project, starting with the observation of existing solid waste management practices, identification of segregation problems, awareness activities and proper use of the existing system, and its review.

Overall, Week 8 completed the field activities of the project and provided an opportunity to review the work carried out during all eight weeks. The project provided practical knowledge about solid waste collection, segregation, storage, and management in Kaikaluru village.

CHAPTER 6
OUTCOMES DESCRIPTION

- Details of the field observations and activities conducted in Kaikaluru village to understand the existing solid waste management practices, segregation methods, and common problems related to solid waste handling. The following observations and outcomes were identified during the project.

1.What are the common types of solid waste observed in the selected areas?

Ans: The commonly observed solid waste was broadly identified as biodegradable and non-biodegradable waste. These include food and vegetable waste, paper, plastic, bottles, and other household waste.

2.How is solid waste collected and stored in the selected areas?

Ans: Solid waste is collected and temporarily stored at different locations before disposal. The field observations helped us understand the existing collection and storage practices.

3. Is biodegradable waste separated from non-biodegradable waste?

Ans: The field observations showed that different types of waste may be mixed during storage and collection. This indicated the need for proper segregation at the source.

4. What problems are caused by mixing different types of solid waste?

Ans: Mixing biodegradable and non-biodegradable waste makes proper collection, recycling, treatment, and disposal more difficult. It also reduces the effectiveness of waste management practices.

5.What approach was identified to improve source segregation?

Ans: Awareness about the existing two-bin system was identified as a practical approach for separating biodegradable and non-biodegradable waste at the source.

6. What was the main outcome of the project?

Ans: Awareness was created about separating wet and dry waste. People were encouraged to use the existing two-bin system properly and avoid mixing waste.

7.What did the project help us learn?

Ans: The project provided practical knowledge about solid waste collection, storage, segregation, and management. It also helped us understand how a simple method can be applied to address a local solid waste management problem.

PROBLEMS IDENTIFICATION

Problems identified in the community:

- Improper Waste Segregation: Biodegradable and non-biodegradable waste are often mixed together during storage and collection, making further waste management difficult.

- Mixing of Wet and Dry Waste: Different types of solid waste are sometimes placed together, which can make the waste handling, recycling, treatment, and disposal processes more difficult.

•Lack of Awareness: Insufficient awareness about separating wet and dry waste can make source segregation difficult for households and other waste-generating areas.

- Improper Waste Storage: Solid waste may remain stored at unsuitable locations before collection, which can affect cleanliness and proper waste handling.

- Difficulties in Waste Collection: Mixed waste makes collection and further handling more difficult because different types of waste need to be separated before appropriate management.

- Limited Source Segregation: Proper segregation at the point where waste is generated is an important requirement for effective solid waste management.

- Improper Disposal Practices: When waste is not properly segregated and managed, it can create difficulties during transportation, treatment, recycling, and final disposal.

PROPOSED SOLUTIONS

Solutions for Plastic Pollution and Waste Management:

- Promote Proper Waste Segregation: Separate biodegradable and non-biodegradable waste at the source to make collection and further waste management easier.

•Promote Proper Use of Existing Bins: Encourage people to use the existing bins for biodegradable and non-biodegradable waste so that different types of waste can be stored separately.

•Promote Awareness About the Existing Two-Bin System: Explain proper use of the existing system One bin for biodegradable waste and another for non-biodegradable waste.

- Improve Waste Storage: Store waste in suitable and separate containers before collection to reduce mixing and make handling easier.

- Encourage Source-Level Segregation: Waste should be separated at the place where it is generated rather than separating it later during collection or disposal.

- Improve Collection and Handling: Properly segregated waste can be collected and handled more systematically according to its type.

- Support Recycling and Proper Disposal: Separating non-biodegradable waste can make further recycling and appropriate disposal easier.

- Maintain the Two-Bin System: Regular observation and proper use of the two-bin system can help maintain source segregation and improve local solid waste management practices.

CHAPTER 7
CONCLUSION OF SOLID WASTE MANAGEMENT

In conclusion, proper solid waste management is essential for maintaining clean and healthy surroundings in the community. Effective management begins with proper segregation of waste at the source, where biodegradable and non-biodegradable waste are separated before collection and disposal.

The Community Service Project carried out in Kaikaluru village helped us understand the existing solid waste management practices and identify problems related to waste collection, storage, segregation, and disposal. The field observations highlighted the importance of reducing the mixing of different types of solid waste and improving source-level segregation.

The project focused on awareness about proper use of the existing two-bin system.
Awareness activities highlighted source segregation.
encouraged proper use of the existing bins at the community level.

Proper segregation makes the further collection, handling, recycling, and disposal of waste easier. The project provided practical knowledge about solid waste management and helped us understand the importance of maintaining proper waste segregation practices.

Overall, the eight-week project provided useful field experience and demonstrated a simple and practical approach to improving solid waste management in Kaikaluru village.

Student Self-Evaluation for the Community Service Project

Student Name V.UMA SRI(24481A0156)
G. CHAKRI SAI(24481A0156)
K.KRISHNA KOWSHIK(24481A0122)

Period of CSP **:** From to

Evaluation:

Name of the Person in-charge: Mrs.U.PALLAVI

Address with mobile number

:

Please rate the your performance in the following areas:

Rating Scale: **1 is lowest and 5 is highest rank**

1)	**Oral communication**	**1**	**2**	**3**	**4**	**5**
2)	**Written communication**	**1**	**2**	**3**	**4**	**5**
3)	**Proactiveness**	**1**	**2**	**3**	**4**	**5**
4)	**Interaction ability with community**	**1**	**2**	**3**	**4**	**5**
5)	**Positive Attitude**	**1**	**2**	**3**	**4**	**5**
6)	**Self-confidence**	**1**	**2**	**3**	**4**	**5**
7)	**Ability to learn**	**1**	**2**	**3**	**4**	**5**
8)	**Work Plan and organization**	**1**	**2**	**3**	**4**	**5**
9)	**Professionalism**	**1**	**2**	**3**	**4**	**5**
10)	**Creativity**	**1**	**2**	**3**	**4**	**5**
11)	**Quality of work done**	**1**	**2**	**3**	**4**	**5**
12)	**Time Management**	**1**	**2**	**3**	**4**	**5**
13)	**Understanding the Community**	**1**	**2**	**3**	**4**	**5**
14)	**Achievement of Desired Outcomes**	**1**	**2**	**3**	**4**	**5**
15)	**Overall Performance**	**1**	**2**	**3**	**4**	**5**

Date: **Signature of the Student**

Evaluation by the Person in-charge in the Community/Habitation

Student Name	**:** V.UMA SRI(24481A0156) G. CHAKRI SAI(24481A0156) K.KRISHNA KOWSHIK(24481A0122)
Period of CSP	**:** From to
Evaluation:	
Name of the Person in-charge:	Mrs.U.PALLAVI
Address with mobile number:	

Date of Please rate the your performance in the following areas:

Rating Scale: **1 is lowest and 5 is highest rank**

1) Oral communication	**1**	**2**	**3**	**4**	**5**
2) Written communication	**1**	**2**	**3**	**4**	**5**
3) Proactiveness	**1**	**2**	**3**	**4**	**5**
4) Interaction ability with community	**1**	**2**	**3**	**4**	**5**
5) Positive Attitude	**1**	**2**	**3**	**4**	**5**
6) Self-confidence	**1**	**2**	**3**	**4**	**5**
7) Ability to learn	**1**	**2**	**3**	**4**	**5**
8) Work Plan and organization	**1**	**2**	**3**	**4**	**5**
9) Professionalism	**1**	**2**	**3**	**4**	**5**
10) Creativity	**1**	**2**	**3**	**4**	**5**
07) Quality of work done	**1**	**2**	**3**	**4**	**5**
12) Time Management	**1**	**2**	**3**	**4**	**5**
13) Understanding the Community	**1**	**2**	**3**	**4**	**5**
14) Achievement of Desired Outcomes	**1**	**2**	**3**	**4**	**5**
15) Overall Performance	**1**	**2**	**3**	**4**	**5**

Date: **Signature of the Supervisor**

Program Outcomes (POs):

Engineering Graduates will be able to:

1.Engineering Knowledge: Apply engineering knowledge and fundamentals to understand and address solid waste management problems in the community.

2. Problem Analysis: Identify and analyze problems related to solid waste collection, storage, segregation, and disposal.

3. Design/Development of Solutions: Develop simple and practical solutions for solid waste segregation by considering environmental and community requirements.

4. Conduct Investigations of Complex Problems: Conduct field observations, collect relevant information, analyze the observed waste management practices, and draw suitable conclusions.

5. Modern Tool Usage: Use appropriate tools and resources for collecting, recording, organizing, and documenting information related to the project.

6. The Engineer and Society: Understand the responsibilities of engineers towards society and apply engineering knowledge to address community-level solid waste management problems.

7. Environment and Sustainability: Understand the environmental impact of improper solid waste management and recognize the importance of sustainable waste segregation and disposal practices.

8. Ethics: Follow ethical principles and responsible practices while carrying out field activities and documenting the project work.

9. Communication: Communicate effectively with team members and the community and prepare clear reports and documentation of the project activities.

10. Project Management and Finance: Apply basic project planning and management principles

while organizing and completing the Community Service Project activities.

11. **Life-long Learning:** Develop the ability to learn from field experiences and understand the need for continuous learning about environmental and sustainable engineering practices.